PREDICTIVE MAINTENANCE SOLUTION FOR SAM AUTOMATION

Viktória Kubišová, Monieka Hardjosoedarmo, Chrislande Duterloo, Bartosz Kudyba and Kajetan Neweś

1 INTRODUCTION

This proposal outlines the development of a **predictive maintenance prototype** to help SAM Automation transition from a **reactive** to a **proactive** service model. The project will leverage **robot sensor data**—such as pressure, vibration, and temperature—to build a **scalable, cross-brand dashboard.**

The prototype will demonstrate how signals from collaborative robots can be collected, analyzed with anomaly detection methods, and translated into **clear, actionable alerts for technicians**. Data may be ingested from robot controllers (via PLC/serial or APIs) and complemented by external sensors where needed.

The expected outcome is a **functional demo by January 2026** that visualizes key signals, provides early warnings of potential failures, and validates both the **technical feasibility** and **user acceptance** of a predictive maintenance system. This will position SAM as a leader in **data-driven maintenance**, delivering value through **improved reliability, reduced downtime, lower costs, and stronger client trust.**

2 PROBLEM STATEMENT AND RESEARCH QUESTION(S)

2.1 Business Problem

SAM Automation provides robotic solutions for packaging in food and pharmaceutical industries, where reliability and minimal downtime are critical. Although modern collaborative robots do generate **internal telemetry** (e.g., joint loads, temperature, fault codes), this data is often limited to **brand-specific controllers**, not consistently recorded, and rarely analyzed for predictive use. As a result, failures are still handled **reactively**, leading to costly disruptions and emergency interventions.

To move toward **predictive** maintenance, SAM needs a system that can make better use of **existing internal signals** while also integrating **external sensors** (e.g., pressure, vibration, temperature) to cover gaps and enable **cross-brand monitoring**. Without such a solution, maintenance teams lack **early warnings** of malfunction, and SAM risks falling behind competitors who are beginning to introduce **predictive platforms**—raising client expectations and shifting industry standards.

2.2 Research Objectives and Questions

This project explores how predictive maintenance can be implemented effectively in SAM's mixed-fleet context. Key questions include:

1. Which internal and external sensor signals best predict failures?
2. What techniques detect anomalies early when failure data is limited?
3. How much lead time is needed for technicians to intervene effectively?
4. What alert formats (traffic-light signals, evidence cards, explanations) prompt response and avoid fatigue?
5. How can heterogeneous robot data be securely integrated into a unified dashboard?
6. What do users expect from the dashboard interface for it to be practical and trustworthy?

2.3 Significance and Urgency

The need to solve this problem goes beyond technical performance. In packaging environments, especially food and pharmaceuticals, downtime has immediate and costly consequences. A predictive platform would increase reliability, reduce costs, and build client trust. With a prototype due by **January 2026**, timely delivery is critical to validate feasibility and position SAM as a leader in data-driven maintenance.

3 Literature Review

3.1 Context and History of the Problem

Industrial maintenance has undergone a clear evolution over the past decades. **Reactive maintenance (RM)**, or run-to-failure, was the earliest widespread strategy: equipment was repaired only after breakdown, maximizing short-term utilization but resulting in high unplanned downtime, collateral damage, and large repair costs (Zhu et al., 2019). **Preventive maintenance (PM)** emerged to address these risks by scheduling interventions at fixed intervals, based on elapsed time or usage cycles (Zhu et al., 2019). While PM reduced catastrophic failures compared to RM, it introduced inefficiencies through unnecessary part replacements and scheduled downtime. Studies by Zhu et al. (2019) and Murtaza et al. (2024) show that **reactive repairs are typically three times more expensive than preventive actions**, yet preventive strategies still lack precision because they are not tied to the actual condition of equipment. **Predictive maintenance (PdM)** was developed as a response, aiming to perform interventions only when sensor data indicate degradation, balancing repair and prevention costs while maintaining system reliability (Zhu et al., 2019).

The economic importance of this transition is evident. Downtime is highly costly: Amazon reported a loss of **USD 4 million during a 49-minute outage** in **2013**, while industry averages estimate losses of **USD 138,000 per hour of downtime** (Zhu et al., 2019). In heavy industry, maintenance expenditures dominate operating costs: operations and maintenance represent **20–35% of revenues** in offshore wind power and **15–70% of production costs** in oil and gas (Zhu et al., 2019). At a macro scale, the International Energy Agency estimates that unplanned outages in energy-intensive sectors cost nearly **USD 50 billion annually**, with facilities losing **3–5% of production capacity** (García et al., 2025).

The technological foundations of PdM emerged from advances in **condition monitoring techniques** such as vibration analysis, thermography, and acoustic monitoring, which enabled the first systematic approaches to fault detection (Zhu et al., 2019). With the onset of **Industry 4.0**, these methods were augmented by the integration of **IoT sensors, big data pipelines, and artificial intelligence algorithms** (Murtaza et al., 2024). IoT devices made it possible to collect continuous, high-frequency operational data, while scalable data infrastructures and analytics transformed raw sensor streams into health indicators and failure predictions (Givnan et al., 2021). This marked the transition of PdM from **isolated condition-monitoring tools to scalable, factory-wide maintenance strategies** (Zhu et al., 2019).

The impact of these developments has been documented across industries. Predictive approaches have been shown to **reduce downtime by up to 50%** and **lower maintenance costs by 10–40%** compared to preventive maintenance (Zhu et al., 2019). A systematic review of implementations from 2017 to 2024 further confirmed that predictive strategies typically cut **downtime by 10–20%** and deliver **global savings of USD 8–15 billion per year** in energy-intensive industries (García et al., 2025). Beyond direct cost reductions, PdM also decreases spare parts inventory, avoids unnecessary interventions, and increases overall equipment availability and lifetime (Murtaza et al., 2024).

Looking forward, the transition to **Industry 5.0** reframes the role of PdM. Whereas Industry 4.0 positioned predictive maintenance primarily as a technical solution for reducing downtime and costs, Industry 5.0 emphasizes **sustainability, resilience, and human-centric design** (Murtaza et al., 2024). This means PdM is now expected not only to optimize schedules and extend equipment life but also to reduce wasted energy from degraded machinery, contribute to resource efficiency, and provide outputs that are interpretable and usable by technicians in **complex, human–machine collaborative environments** (García et al., 2025).

3.2 Review of Existing Solutions, Competitors, and Best Practices

Predictive maintenance solutions are often built on standardized frameworks. **ISO 13374** defines the **Open System Architecture for Condition-Based Monitoring (OSA-CBM)**, which specifies six functional blocks: data acquisition, data manipulation, state detection, health assessment, prognostics, and advisory generation (Zhu et al., 2019). PdM 4.0 architectures expand this model by adding preprocessing, decision support, and maintenance execution modules, integrating IoT devices, cyber-physical systems, and cloud computing for high-frequency, factory-wide deployments (García et al., 2025; Murtaza et al., 2024).

Signal processing remains central to PdM, with **Fast Fourier Transform (FFT), wavelets, and CEEMDAN** widely used to extract features from noisy time-series data. These techniques are combined with machine learning models such as **SVMs, CNNs, and LSTMs** for **fault detection** and **Remaining Useful Life (RUL) estimation**. Hybrid models that merge physics-based and data-driven approaches improve robustness under non-stationary conditions and increase interpretability (García et al., 2025).

In robotics, Morettini (2021) demonstrated PdM using only internal sensors. Torque profiles were converted into a joint health index, and anomalies were predicted with **Gaussian Process Regression, Exponentron, and hybrid models**, with hybrid methods performing best. The study also noted reduced accuracy with low sampling rates or missing data. Eang and Lee (2025) proposed a **CNN–RNN hybrid observer** for DC motor drives, analyzing temperature and rotational speed to detect gear tooth wear and misalignments. Their model outperformed **CNN–LSTM** in both accuracy and computational efficiency, supporting real-time deployment. Wojtulewicz and Chaber (2025) integrated PdM directly into robot controllers via **IIoT**. Process data from CNC and robot controllers were analyzed in **PLCs** to calculate component consumption indicators, visualized on HMI panels, and shared via MQTT, achieving predictive diagnostics without additional sensors.

Visualization and interpretability are also advancing. Paraschos et al. (2021) developed **VisioRed**, a tool that combines **PCA-based dimensionality reduction with explanation techniques (LIME, LioNets)** to show feature importance in time-series PdM models. It also includes a prescriptive module (XYZ) that recommends sensor value changes to extend RUL.

Real-time anomaly detection has been advanced by approaches using deep learning, in particular **stacked autoencoders (SAEs).** Givnan et al. (2021) proposed a real-time PdM framework where SAEs are trained on **healthy operational data** to reconstruct input signals; anomalies are detected as deviations measured by **reconstruction error**. Their system introduced a **flexible traffic-light scale** (green for normal, amber for concerning, red for critical) that allows early intervention before breakdown. The framework was designed for edge deployment to **minimize bandwidth and storage** by transmitting only **anomaly-related windows**, making it practical for industrial use cases with high-frequency sensor data.

Industrial competitors such as **ABB, FANUC, and KUKA** embed PdM features in **proprietary platforms** (e.g., ABB Ability, FANUC Zero Downtime, KUKA Connect). Bogue (2025) notes that while these solutions deliver value within their ecosystems, they are **brand-specific**. Independent providers, such as SAM Automation, have an opportunity to differentiate by building vendor-agnostic platforms that unify heterogeneous robot data while adopting best practices from academic and industrial advances.

3.3 Gaps in Current Knowledge or Approaches

Despite significant progress, several limitations remain in predictive maintenance research and practice. A major challenge is **data scarcity and imbalance**: industrial datasets contain abundant normal operation data but few fault events, forcing reliance on simulations or lab experiments (García et al., 2025; Zhu et al., 2019). Models must also contend with non-stationary environments, where varying loads, speeds, and noise complicate feature extraction and reduce reliability (García et al., 2025).

Another gap lies in **interpretability**. Deep learning models achieve high accuracy but are often black boxes; tools such as VisioRed provide transparency but are not yet widely adopted in practice (Paraschos et al., 2021). **Integration** challenges persist as well: most industrial PdM platforms (e.g., ABB, FANUC, KUKA) are proprietary and vendor-specific, limiting their applicability in multi-brand environments (Bogue, 2025).

Finally, there is a **deployment gap** between laboratory studies and industrial reality. Many solutions are validated on **controlled datasets** (e.g., NASA turbofan, simulated torque wear) but perform less reliably with missing data, low sampling rates, or fluctuating operating conditions in factories (Givnan et al., 2021; Morettini, 2021). In the shift toward Industry 5.0, PdM must also address sustainability and human-centricity, reducing energy waste and delivering interpretable outputs that technicians can act upon (Murtaza et al., 2024).

To help address these gaps, this project aims to design a **cross-brand, multi-machine predictive maintenance platform**. By combining anomaly detection techniques with robust data pipelines, the system will seek to capture subtle patterns and nuances in robot behavior across heterogeneous environments, moving toward a vendor-agnostic and more resilient PdM solution.

4 RESEARCH OBJECTIVES

The immediate goal of this project is to produce a **working prototype** that reliably ingests robot signals, with a primary focus on **suction tooling pressure and vacuum levels**, complemented by **vibration** and **temperature** where feasible.

Data will be collected either from robot controllers, using **PLC or serial connections** (and **REST APIs** where available), or from **external sensors** added to fill gaps and support **cross-brand applicability**. These signals will be visualized in an **interactive dashboard** designed to provide technicians with an accessible view of robot health.

Within this prototype, the team will demonstrate that **sensor data** can be transformed into **actionable maintenance insights** through both **threshold-based baselines** and **anomaly detection** using **stacked autoencoders**. The system will visualize signal trends, issue clear **traffic-light alerts**, and provide short **evidence cards** to help technicians interpret anomalies without requiring advanced data expertise. A key part of the project is to capture structured feedback from **SAM technicians** and **client-side staff**, ensuring that the outputs are not only technically correct but also understandable and useful in practice.

The primary objective is to validate the **feasibility** of moving SAM from **reactive** to **proactive maintenance** by showing that raw robot signals can provide early, **actionable warnings** of tooling failures. Secondary objectives include comparing a **log-only setup** with a **live inference mode**, identifying which signals or combinations give the most reliable indicators of failure, and shaping a **dashboard design** that offers the necessary depth for engineers while remaining intuitive for technicians. The project will also document **data limitations**, such as the scarcity of **historical failures** and reliance on **simulated fault scenarios**, and evaluate their impact on predictive accuracy while outlining clear next steps for scaling to client robots.

Success will be measured by the delivery of a **functional demo** that both visualizes robot signals and generates meaningful alerts, by the system's ability to detect at least **two simulated failure modes** with acceptable precision (targeting at least **70 percent precision for red alerts** and **60 percent recall overall**), and by positive **usability feedback** from technicians, with a target rating of **4 out of 5 or higher**. Finally, the project will produce a **decision brief** that summarizes the prototype's limitations, proposes a **prioritized roadmap**, and provides SAM leadership with clear criteria for deciding on further investment and potential deployment.

5 Methodology

This project adopts a **mixed-methods approach**, combining **quantitative analysis** of sensor data with **qualitative validation** through controlled failure simulations. Data will be collected from both **external and internal sources**, with a primary focus on **pressure and vacuum signals**, complemented by **vibration** and **temperature sensors**, as well as robotic arm sensors measuring **position**, **temperature**, and **voltage**. To create realistic conditions for anomaly detection, we will mimic **failure scenarios** in the lab and label the resulting data for model training.

The concept model will utilize a **Niryo Ned 2 robot** equipped with a **vacuum pump** and a single **suction cup**. Since this default end effector is optimized for flat surfaces and does not typically reflect client use cases, we will try to approximate realistic conditions by providing **cardboard boxes**, adjusting their weight, and examining the surface characteristics. The process of data collection will involve the repetitive task of **moving boxes** from one place to another, ensuring consistent and varied interactions for comprehensive data analysis. To capture changes in behavior, externally attached **sensors** will be connected either directly to a **laptop** via **PLC/serial** or **API interfaces**, or through a **Raspberry Pi** unit integrated into the robot's base. The **Raspberry Pi** serves as a **flexible bridge** for synchronizing external sensor streams with internal robot signals, complementing the **built-in telemetry** and enabling a broader analysis of potential failure causes, particularly in the client's primary area of concern: **pick-and-place tooling failures**. In this setup, the **Raspberry Pi** is used selectively to assist with **data acquisition** or **robot steering** when required, while the **laptop** remains the **primary processing unit** for **modeling** and **dashboard integration**.

Raw **sensor data** will be **cleaned**, **preprocessed**, and segmented into interval windows suitable for **sequential modeling**. Using an **LSTM autoencoder**, the system will learn to reconstruct typical operational patterns. **Reconstruction errors** between true and predicted values will then be computed and compared against **adaptive thresholds** to detect **anomalies**. Divergence levels will be classified into three categories—**green** (normal behavior), **amber** (warning), and **red** (critical abnormality)—to facilitate intuitive monitoring. Signals of anomalous incidents will be transmitted to a **user-friendly dashboard**, where **alerts** will be displayed alongside **visualizations** of the anomaly for quick interpretation and decision-making.

The workflow will follow a structured sequence: initial setup of **data acquisition** (via **laptop** or **Raspberry Pi**, and **sensors**), **exploratory data analysis (EDA)** across test scenarios, **data collection** of both healthy and faulty runs through **controlled breakdown simulations**, **data preparation** and **local modeling**, and integration of **anomaly detection** into a **real-time dashboard**. **Validation** will be conducted by reusing the solution on a **second robotic arm** as a **proof of concept**, demonstrating **cross-brand generalizability**, with rigorous **testing** to assess **accuracy**, **robustness**, and **scalability**, and **technician feedback** to evaluate **usability** and **alert effectiveness**.

Two operational modes will be compared during testing: **Option A (log-only mode)**, where data is collected and analyzed offline, and **Option B (live inference mode)**, where anomaly detection runs in real time on the **laptop**. In later phases, **cloud-based storage** and **retraining pipelines** may be explored to support **scalability** and ongoing **model improvement**, though this remains beyond the immediate **MVP scope**.

6 Expected Outcomes & Deliverables

The project will deliver a **predictive maintenance prototype** capable of detecting **anomalies in real time** and translating them into clear, actionable insights through an **interactive dashboard**. The prototype will ingest signals from the **Niryo Ned 2 robot**—with a primary focus on pressure and vacuum, supported by vibration, temperature, and built-in telemetry—and transform them into **traffic-light alerts** and evidence cards that technicians can interpret quickly.

Key deliverables include a working **real-time dashboard (Streamlit)** integrated with live robot data; **healthy and faulty loop datasets** from controlled breakdown simulations, labeled, archived, and documented for evaluation and future retraining; **baseline methods** alongside a **stacked autoencoder model** with **signal trend plots**, **reconstruction error visualizations**, and **anomaly alerts**; and **validation on a second robot** as a **proof of concept** for **cross-brand generalizability**. **Technician testing** will provide structured usability feedback, ensuring the system is both technically robust and practical in operation.

Success will be measured by the system's ability to detect at least **two simulated failure modes** with **meaningful precision**, by achieving a **usability rating of 4/5 or higher** from technician testing, and by delivering a **prototype presentation** and **pilot reports** that demonstrate feasibility, scalability, and a clear **roadmap for scale-up**.

While the **MVP** focuses on **local, laptop-based inference**, the groundwork will also be laid for **cloud storage** and **retraining pipelines** as future extensions, ensuring the solution can evolve into a **vendor-agnostic, scalable platform** for predictive maintenance. The expected impact extends beyond this **proof of concept**, equipping **SAM** with a validated **foundation for predictive maintenance** and a clear decision basis for wider investment and deployment.

7 Timeline & Milestones

The 18-week project follows the **CRISP-ML(Q) cycle** and is structured around **five milestone presentations** (Weeks 4, 8, 10, 14, and the final launch in Week 18). Each milestone is preceded by a **Datalab dry-run** to refine quality and delivery. Bi-weekly feedback sessions (Weeks 2–16) and showcases (Weeks 3–17) ensure continuous alignment and agile adjustments.

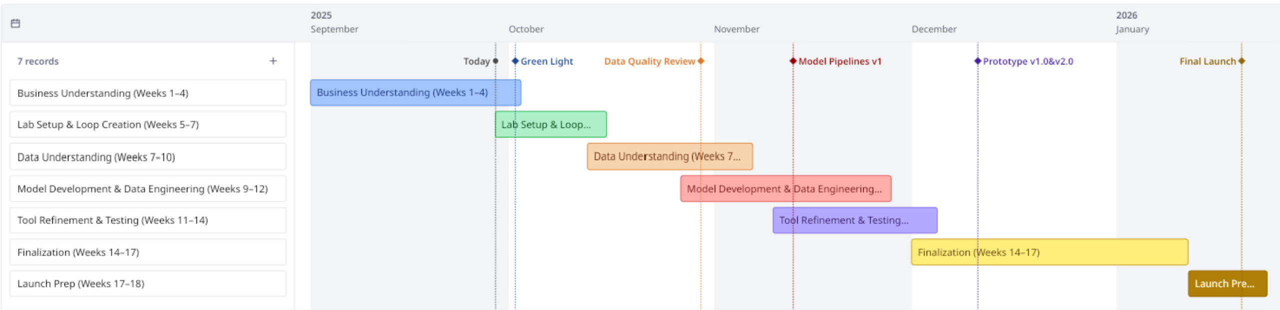

Figure 1. *High-level project timeline with CRISP-ML(Q) phases and milestones. Detailed descriptions of each phase and deliverable are provided below.*

Weeks 1–4 (Business Understanding): Problem analysis, project scoping, risk assessment, and literature review. Deliverables include the Business Requirements Document (BRD), a risk register, and the formal project proposal. **Milestone M1** (end Week 4): Green Light presentation and approval to proceed.

Weeks 5–7 (Lab Setup & Loop Creation): Instrumentation of the first university robot, delivery of a tool/sensor list, and definition of operational loops. The first robot loop will be implemented, with test runs documented. Deliverables include a lab setup report and procedures for accessing/logging robot and sensor data.

Weeks 7–10 (Data Understanding): Collection of healthy and faulty loop datasets, with simulated faults and event labeling. Exploratory Data Analysis will define normal ranges, draft baseline thresholds, and produce the first Data Quality Report. **Milestone M2** (end Week 8): Data Quality Review and confirmed signal set.

Weeks 9–12 (Model Development & Data Engineering): Development of ingestion pipelines and feature engineering, training of anomaly detection baselines (thresholds, Isolation Forest, SAE), and generation of reconstruction error plots. **Milestone M3** (end Week 10): Model cards and pipelines v1.

Weeks 11–14 (Tool Refinement & Initial Testing): Integration of a technician-facing dashboard (Streamlit MVP, Option A log-only) with live signals and alert visualizations. Pilot #1 will run on the Niryo robot; Pilot #2 will extend testing to the UFactory robot and validate PLC integration. Deliverables include an early dashboard version and pilot reports. **Milestone M4** (Week 14): Prototype v1.0 presentation across both lab robots.

Weeks 14–17 (Finalization): Refinement of thresholds and anomaly logic, addition of live inference mode (Option B), and usability testing with technicians (target SUS $\geq$ 4/5).

Weeks 17–18 (Launch Preparation & Scale-Up): Packaging of datasets, code, and documentation, along with delivery of a Scale-Up Plan outlining how the prototype could be extended to client robots. **Milestone M5** (Week 18): Final stakeholder presentation on 22/01.

Week 18 (Final Launch): Submission of all deliverables on 23/01 (17:00).

7.1 Critical Dependencies

Project success depends on the capabilities of the **university robots**, especially whether they provide usable readings for **suction tooling pressure** and, where possible, **vibration** and **temperature**. If these internal signals are unavailable or insufficient, **external sensors** may need to be added, which could increase setup complexity and integration time.

Another dependency is how well the **lab environment** can reproduce fault scenarios that resemble those in SAM's client factories. Without **historical failure data**, the project relies on **simulated or induced faults**, and the accuracy of modeling depends on how realistic these simulations are and how well they can be **labeled**.

The reliability of **anomaly detection** will also depend on **data coverage and integrity**. Low sampling rates, missing timestamps, or sparse fault events could weaken model performance. To mitigate this, the project will begin with **rule- and threshold-based alerts** before moving to **ML-based anomaly detection** as more data becomes available.

Looking ahead, **scaling beyond the lab** would require **secure API access** to SAM's robots and close **IT-OT collaboration** to ensure safe data flows. While this lies outside the **MVP's primary scope**, it remains relevant for potential validation with SAM's equipment if time permits.

8 RESOURCES & BUDGET

8.1 People and time

The prototype can be delivered by the student team specified for this project – **one data scientist, one data analyst, two analytics translators, and one data engineer** – working in close partnership with SAM Automation's focal point (Stan Jacobs) and a small group of service/maintenance technicians for hands-on validation. The analytics translators steer requirements and success criteria, the data engineer owns ingestion from **PLC/serial connections (and APIs where available)** and log storage, and the data scientist with the analyst build signal dashboards and first-pass alerting. Expect one brief review per week with the SME and technicians for feedback on thresholds, usability, and false positives, plus light academic oversight from the BUas supervisor to keep milestones on track.

8.2 Infrastructure, tools, and environments

For data handling, the team will stand up a minimal pipeline that reads robot signals (e.g., **pressure, vacuum, vibration, temperature, motor load, fault codes**) through **PLC/serial connections and APIs where available** and lands them in a small store (files or a lightweight database) suitable for time-series visualization. A lightweight analytics environment (**Python**) and a dashboard layer (**Streamlit for the MVP**, with Power BI as a possible future extension) will host the prototype. To respect operational constraints, processing will favor a **laptop-first setup**, with the option of moving to **live inference** later. Read-only access, tokenized authentication, and basic observability (ingestion health, latency, and data quality checks) will be ensured. Version control, issue tracking, and reproducible builds (**Docker**) round out the delivery setup so that the prototype is demo-ready but remains clearly non-production.

8.3 Data access, governance, and security

Access to the demo robots' **signals**—whether exposed via **PLC/serial connections, APIs (e.g., REST) where available, or external sensors**—is the principal dependency. The project will operate against a **test or mirrored environment** where possible, use the **narrowest scopes** and **least-privilege credentials**, and **log access** for auditability. Data will be **minimized** to only the signals required for dashboards and alerts, retained only for the **prototype window**, and **labeled with provenance and timing** to aid back-testing. Recommendations will be presented as **decision support**, not control actions, and will link to existing **SOPs** to avoid unsafe autonomy.

8.4 Indicative budget model (prototype phase)

Provided by University (no additional cost)

- Azure environment (compute, storage, networking) – **€0***
- Data hosting/backup in university Azure – **€0***
- Robot arms + OEM software/licenses – **€0***
- Lab access & support – **€0***

**Assumes sufficient credits/quotas during the MVP window; internal policies apply (read-only use, no production SLAs, limited external access).*

Cash costs (only if not already covered)

- Dashboard licensing (e.g., a few Power BI Pro seats, only if Streamlit is not sufficient) – **€0 to €600**
- Optional local gateway/mini PC or Raspberry Pi (to aggregate on-robot data; stretch goal) – **€0 to €1,500**
- Cloud overage outside university credits (only if limits are hit) – **€0 to €400**
- Contingency (around 15%) applied to the above

Totals (indicative)

- Minimal path (all university coverage, open-source tools with Streamlit): **€200–€900**
- With optional gateway + paid seats (and possible small overage): **€1,200–€3,800**

Excludes internal staff time and any production-grade services. Prices are VAT-exclusive.

Sensitivity (if coverage changes)

If university credits/equipment become unavailable or limits are exceeded, expect **€100–€300/month** for basic compute + storage and **€400–€1,500** one-off for a gateway, depending on final architecture.

8.5 Risk Assessment

#	Risk	L	I	Owner	Early indicators (watch)	Mitigation (now)	Contingency (if triggered)
1	Weak predictions due to limited failure historical data	High	High	DS Lead	Sparse labels; model lifts $<$ baseline; high false negatives	Start with condition-based & anomaly baselines; physics/threshold rules per signal (temp/load/fault codes); define precision/recall targets per alert tier	Reframe as advisory early-warning; narrow scope to top 2–3 failure modes; extend data collection period
2	Data quality & availability (sensors/PLCs)	Med	High	DE Lead / IT-OT	Missing timestamps; irregular sampling; PLC/serial dropouts	Local buffering + retries; schema contracts; data validation checks; drop/flag rules; observability dashboard	Switch to batch ingestion with backfills; reduce signal set to the most reliable channels
3	Local (on-robot) vs cloud trade-off misfit	Med	Med	IT-OT / Architect	Latency spikes; offline periods; cost overruns	Pilot both: minimal local (on-robot) summary → cloud storage; data minimization; configurable sampling	Offline mode: local caching + deferred upload; reduce telemetry frequency
4	Alert fatigue / UX rejection	Med	High	Analytics Translator	Technicians mute alerts; tickets ignored	Severity tiers; per-robot baselines; rate-limit & snooze; "Why this alert?" explainers; feedback button	Temporarily raise thresholds; focus on 2–3 high-value alerts; weekly tuning with field input
5	Unsafe or over-confident recommendations	Low	High	Product Owner / QA	Users ask, "Can I act without SOP?"	"Decision-support only" wording; link to SOPs; require human acknowledgment; show confidence & evidence	Disable auto-recs; keep signals + guidance only until validated
6	Security/OT exposure of robot data	Med	High	IT-Security / IT-OT	Unapproved ports; shared credentials; missing audit	Least-privilege tokens; rotation; network segmentation; audit logs	Cut external access; read-only scope; security review before any expansion
7	Stakeholder availability (SME time)	Med	Med	Product Owner	Missed reviews; unclear acceptance criteria	Lock weekly 30-min review; RASCI; written acceptance tests for alerts & dashboard	Trim backlog; defer lower-value features; escalate blockers early
8	Scope creep to production (explicitly out of scope)	Med	Med	Product Owner / PM	Requests for factory integration, SLAs	Reiterate scope in every review; MoSCoW; DoD = prototype + demo	Park prod items in "Phase 2" log; no code to live networks
9	Model drift / task changes on demo arm	Med	Med	DS Lead	Gradual precision drop; new false positives	Time-split validation; drift monitors; versioned configs	Roll back to prior thresholds; re-baseline per task
10	Tooling friction (dashboard choice limits)	Low	Med	DE Lead	IT says "no" to selected tool	Choose one path (e.g., Streamlit or Power BI) early; package in Docker	Export static demo build; switch to alternative tool with same data layer

Table 1. *Risk register with likelihood (L), impact (I), owners, and mitigation strategies.*

9 CONCLUSION

This proposal has outlined a strategic project to develop a **predictive maintenance prototype** for SAM Automation's robotic fleet. The initiative directly addresses the critical business problem of **unplanned downtime** by transitioning the company from a **reactive** to a **data-driven, proactive maintenance model**. By leveraging **existing robot signals and complementary external sensors** through a secure, cross-brand dashboard, the solution will provide **early anomaly detection** and **actionable alerts** for technicians.

The proposed plan is **feasible, cost-effective, and designed for timely execution**. The successful delivery of a **functional prototype** will validate the technical approach, demonstrate clear value to clients, and position SAM as a leader in **innovative service solutions**. This project represents an important step towards **improving operational reliability**, **reducing costs**, and **maintaining a competitive edge** in the packaging industry.

The immediate action is for SAM to **grant Green Light approval**, enabling the team to begin the first phase of development with access to the required **robot signals and lab setup**. With a prototype due by **January 2026**, timely approval is essential to secure SAM's position at the forefront of **robotic maintenance innovation**.

DISCLOSURE

This proposal was written with the assistance of *ChatGPT* (OpenAI, GPT-5, 2025). The tool was used only to improve grammar, phrasing, and style. All ideas, analysis, and conclusions are those of the authors.

References

Bogue, R. (2025). Predictive maintenance for industrial robots. *Industrial Robot: The International Journal of Robotics Research and Application*, *52*(3), 305–311. https://doi.org/10.1108/IR-03-2025-0081

Eang, C., & Lee, J.-M. (2025). Comparison of deep learning models for predictive maintenance of industrial robot motor drives. *Sensors*, *25*(1), 25. https://doi.org/10.3390/s25010025

García, R., Fidalgo, E., Alegre, E., Fernández-Robles, L., & Fernández, D. (2025). Condition monitoring and predictive maintenance in industrial equipment: An nlp-assisted review of signal processing, hybrid models and implementation challenges. *Applied Sciences*, *15*(10), 5465. https://doi.org/10.3390/app15105465

Givnan, S., Chalmers, C., Fergus, P., Ortega, S., & Whalley, T. (2021). Real-time predictive maintenance using autoencoder reconstruction and anomaly detection. *arXiv preprint arXiv:2110.01447*. https://arxiv.org/abs/2110.01447

Morettini, M. (2021). *Machine learning in predictive maintenance of industrial robots* [Master's Thesis]. Uppsala University. http://urn.kb.se/resolve?urn=urn:nbn:se:uu:diva-452321

Murtaza, M., da Silva, R., Dall'Oglio, A., et al. (2024). Paradigm shift for predictive maintenance and condition monitoring from industry 4.0 to industry 5.0. *Journal of Industrial Information Integration*, *34*, 100567.

Paraschos, P., Angelopoulos, C. M., Filios, G., & Tserpes, K. (2021). Visiored: A visualisation tool for interpretable predictive maintenance. *arXiv preprint arXiv:2103.17003*. https://arxiv.org/abs/2103.17003

Wojtulewicz, A., & Chaber, P. (2025). Industrial robot control system with a predictive maintenance module using iiot technology. *Sensors*, *25*(4), 1154. https://doi.org/10.3390/s25041154

Zhu, J., Djurdjanovic, D., Qiu, H., Lee, J., Ni, J., & Liao, H. (2019). A survey of predictive maintenance: Systems, purposes and approaches. *IISE Transactions*, *51*(11), 1190–1206.